

Stage 1 — Variables, Loops & Conditionals (1–10)

These build programming confidence.

1. Print Hello World
2. Sum of two numbers
3. Swap two numbers
4. Check Even or Odd
5. Find Largest of three numbers
6. Leap Year
7. Reverse a number
8. Palindrome Number
9. Count digits in a number
10. Sum of digits

Skills:

- if-else
 - loops
 - modulo
 - integer division
-

Stage 2 — Number Theory (11–20)

11. Factorial
12. Fibonacci Series
13. Check Prime
14. Print Prime Numbers till N
15. Count Factors
16. GCD (Euclid Algorithm)
17. LCM
18. Armstrong Number
19. Perfect Number
20. Power of Number (without pow())

Skills:

- nested loops
 - optimization
 - mathematics
-

Stage 3 — Pattern Printing (21–30)

Everyone skips these.

Don't.

They teach loops better than anything.

21.

```
*  
**  
***  
****
```

22.

```
*****  
****  
***  
**  
*
```

23.

```
1  
12  
123  
1234
```

24.

```
A  
AB  
ABC
```

25.

```
1  
22  
333  
4444
```

26.

```
*  
***  
*****  
*****
```

27.

```
*****  
*****
```

*

- 28. Hollow Rectangle
- 29. Hollow Pyramid
- 30. Butterfly Pattern

Skills:

- nested loops
 - visualization
 - debugging
-

Stage 4 — Arrays (31–40)

Now real DSA begins.

- 31. Find Largest Element
- 32. Find Smallest Element
- 33. Sum of Array
- 34. Average of Array
- 35. Linear Search
- 36. Reverse an Array
- 37. Find Second Largest
- 38. Count Even & Odd Numbers
- 39. Remove Duplicates (Sorted Array)
- 40. Rotate Array by One Position

Skills:

- indexing
 - traversal
 - array manipulation
-

Stage 5 — Strings (41–45)

- 41. Reverse String
- 42. Palindrome String
- 43. Count Vowels & Consonants
- 44. Frequency of Characters
- 45. Remove Spaces

Skills:

- characters
 - ASCII
 - traversal
-

Stage 6 — Basic Searching & Sorting (46–50)

- 46. Binary Search
- 47. Bubble Sort
- 48. Selection Sort
- 49. Insertion Sort
- 50. Merge Two Sorted Arrays

Skills:

- searching
 - sorting
 - algorithm thinking
-

After These 50 Problems

You should comfortably solve many beginner problems such as:

- Two Sum
- Valid Palindrome
- Remove Duplicates
- Binary Search
- Maximum Subarray
- Merge Sorted Array
- Best Time to Buy and Sell Stock
- Move Zeroes
- Contains Duplicate
- Plus One

These will feel much more approachable because you've already practiced the underlying techniques.

What You Will Learn

Stage	Concepts
1	Variables, loops, conditions
2	Mathematical logic
3	Nested loops and debugging
4	Arrays and indexing
5	Strings and character manipulation
6	Searching and sorting fundamentals

Study Routine (Recommended)

For each problem:

1. Understand the problem statement.
2. Solve it on paper with a small example.
3. Write the code from scratch (without copying).
4. Dry-run it line by line.
5. Analyze its time and space complexity.
6. Rewrite it after a day from memory.

Aim to solve **2–4 problems per day**. At that pace, you'll finish this foundation in about **2–3 weeks**. Once these problems become comfortable, you'll have the confidence and skills needed to start tackling LeetCode systematically rather than feeling overwhelmed.